Pair blocks are incompatible with the critical head

Human Compression project: L044 continuation

Retain the complete critical-head hypotheses: $n = 2r \geq 18$, $G \leq \mathrm{Alt}(\Omega)$ is transitive with connected string C-marking, and every full ambient maximal omission is intransitive. The acquired six-point head has support D. Write

$$E = \Omega \setminus D, \quad B = G_{\geq 4}, \quad K = G_{\geq 5}, \quad \sigma = \rho_4, \quad x = b^\sigma \neq b.$$

The acquired early-cut calculation gives an odd involution δ supported on E, fixing b, x and centralizing K. The group B fixes D pointwise, and K fixes b.

Lemma 1 (Two alternatives at rank two below the degree)**.** *Let $H = \langle h_1, \ldots, h_{N-2} \rangle$ be a faithful string C-group on $N \geq 5$ points, with connected noncommutation diagram. Suppose deleting each mark changes its orbit partition. If H is transitive, its centralizer in S_N is trivial. If it is intransitive, it fixes one point and is natural S_{N-1} on the rest; its marks are the single edge transpositions of a path, in path order or reverse order.*

Proof. In the transitive case apply the complete accepted L047 theorem. For odd N, rank $N - 2 > N/2$ and all its proper exceptions have even degree. For even $N \geq 8$, $N - 2 > N/2 + 1$, greater than every proper exception rank. Thus in these degrees H is natural A_N or S_N, whose centralizer is trivial. When $N = 6$, the proper rank-four imprimitive exceptions have types, up to reversal,

$$(3, 2, 3), \ (2, 3, 3), \ (2, 3, 4), \ (2, 4, 3).$$

All have disconnected noncommutation diagrams. The only connected proper possibility is the primitive degree-six S_5, also with trivial centralizer. Indeed, in a faithful primitive nonregular action a point stabilizer is maximal and self-normalizing: its normalizer is either itself or the whole group, and the latter would make it normal and then trivial by faithfulness. A centralizing permutation is determined by the image of a point, and the possible images correspond to the normalizer quotient. This proves the asserted triviality in every transitive case.

If H is intransitive, choose for each mark an edge joining different orbits of its omission. The $N-2$ selected edges form a forest, since a cycle would supply an avoiding path. There are exactly two forest components; they are precisely the two H-orbits, since H is intransitive. Deleting a selected edge j splits its tree into two pieces. The actual omission contains every other selected edge and cannot reconnect these pieces, so every other mark preserves that cut. Within a tree the only two distinct vertices not separated by a cut other than cut i are the endpoints of edge i. Thus each h_i is exactly its selected transposition. Each tree supplies its full natural symmetric group. If both trees had edges, their two nonempty sets of marks would commute, disconnecting the diagram. One tree is therefore a singleton. Finally, a vertex of degree three in the other tree would give a triangle of noncommutation among its incident edge transpositions, impossible in a string. That tree is a path, and its connected string ordering is its path order or its reverse. □

Theorem 2. *Under the standing hypotheses, B cannot be transitive on E and preserve a partition into pairs. This covers every critical degree $n \geq 18$ and uses no alternating rank induction.*

Proof. Suppose there are $m = (n - 6)/2 = r - 3 \geq 6$ pairs. The rank of B is $d = r - 4 = m - 1$. Each distinguished omission in B is intransitive on E: otherwise adjoining the head and ρ_3, which links the head to b, would make the corresponding full ambient omission transitive.

Let M be a minimal distinguished subset generating the whole action on the pairs, and let c count the actual point orbits of $\langle M\rangle$ on E. Transitivity on the pairs makes every point orbit meet every pair, so $c \le 2$. The acquired actual-orbit forest and Whiston bound give

$$d - |M| \le c - 1 \le 1, \qquad |M| \le m - 1.$$

Minimality makes the block images of M independent. All marks other than σ fix the pair containing b. Hence $\sigma \in M$ and moves that pair. Let b' be its other point; then x is outside $\{b, b'\}$.

If $|M| = m - 1$, all distinguished block images are independent. The $m - 2$ images from K fix the pair $\{b, b'\}$ and attain maximum independent length on the other $m - 1$ pairs. Whiston's equality clause makes their image natural S_{m-1}. The group K fixes b, b' pointwise. On the other points it has either one transitive orbit or two orbits, each meeting each pair once. In the first case, centralization and fixation of x make δ fix the whole orbit. In the second, the two row actions are faithful natural S_{m-1} actions; δ fixes the row containing x pointwise, and the other row is fixed by its trivial natural centralizer. The remaining fixed set is $\{b, b'\}$; fixation of b forces fixation of b'. Thus $\delta = 1$, a contradiction.

We may therefore suppose $|M| = m - 2$. The forest inequality forces $c = 2$. Let U, V be the two point orbits, with $b \in U$ and $b' \in V$. Each meets every pair once, so either row action of $\langle M\rangle$ is faithful: an element fixing one row fixes every pair and preserves both rows, hence fixes the other row as well. It inherits full IP and all $m - 2$ nonidentity distinguished marks.

Its noncommutation diagram is connected. A component not containing σ fixes b and commutes with the other components. Transitivity then makes it fix the entire row, contrary to faithfulness. Since M contains the first mark of B and omits just one mark, connectedness forces it to omit the last mark: an internal missing index would separate two nonempty commuting intervals. Thus

$$M = \{\rho_4, \dots, \rho_{r-2}\}, \qquad H = \langle M \setminus \{\sigma\}\rangle = G_{5,\dots,r-2}.$$

The group H fixes b, b' and acts faithfully and diagonally on $U \setminus \{b\}$ and $V \setminus \{b'\}$, each of degree $N = m - 1 \ge 5$, with rank $m - 3 = N - 2$. Its diagram is connected, being an interval of the connected marking of M.

Every row omission in H changes its orbit partition. Otherwise the identical row partitions on both sides, with b, b', D fixed, would leave the actual point-orbit partition of H unchanged. Adjoining all the other ambient marks would give the same generated orbit equivalence relation with or without the deleted mark. The full ambient omission would be transitive, a contradiction. Thus Lemma 1 applies to the faithful row marking of H.

If the row action of H is transitive, δ centralizes it and fixes $x \in U \setminus \{b\}$; here x lies in U because $\sigma \in \langle M\rangle$ preserves U. It fixes that whole row, and its action on the other row is trivial by Lemma 1. The fixed points b, b' are fixed as above, again giving $\delta = 1$.

If the row action is intransitive, write p for its unique singleton in $U \setminus \{b\}$ and $W = U \setminus \{b, p\}$. On W, of size $m - 2 \ge 4$, its marks are the transpositions of a path. Let $e = \rho_5$ be its first edge and let a be the endpoint detached by deleting e. Then

$$\mathrm{Fix}_U(G_{6,\dots,r-2}) = \{b, p, a\}.$$

The mark σ commutes with this entire subgroup and therefore preserves that three-point fixed set. On the other hand $\langle M\rangle = \langle \sigma, H\rangle$ is transitive on U, while H fixes both b, p. Since σ is an involution, transitivity forces it to send both b and p into W: a fixed point or an interchange $b \leftrightarrow p$ would leave a singleton or the pair $\{b, p\}$ invariant. Both images would consequently have to equal a, impossible for a permutation. This excludes the last case. □

Attribution and boundary. The critical-head calculation and actual-orbit forest/Whiston interfaces remain acquired. The full accepted L047 theorem, including the exact small exception list and its common theory, finite certificate, software and distinct reviews, remains paid. The new step applies it to the endpoint omission on each literal row; a saturated forest then excludes the intransitive alternative. No B5 classification, smaller-degree alternating induction, pending packet or new computation is used. Other block systems, opposite heads, the unshared endpoint and the complete L044 recovery remain separate; no whole-original or question credit is claimed.